

**IDENTIFYING INFORMATION:**

NAME: Bruce, Juliette

ORCID iD: <https://orcid.org/0000-0001-9766-7896>

POSITION TITLE: Assistant Professor

**PRIMARY ORGANIZATION AND LOCATION:** Dartmouth College, Mathematics, Hanover, New Hampshire, United States**Professional Preparation:**

| ORGANIZATION AND LOCATION                                                     | DEGREE<br>(if applicable) | RECEIPT DATE      | FIELD OF STUDY                                              |
|-------------------------------------------------------------------------------|---------------------------|-------------------|-------------------------------------------------------------|
| Brown University, Providence, Rhode Island, United States                     | Postdoctoral Fellow       | 08/2022 - 06/2024 | Mathematics                                                 |
| University of California, Berkeley, Berkeley, California, United States       | Postdoctoral Fellow       | 08/2020 - 07/2022 | Mathematics                                                 |
| Mathematical Sciences Research Institute, Berkeley, California, United States | Postdoctoral Fellow       | 08/2020 - 05/2021 | Mathematics                                                 |
| University of Wisconsin–Madison, Madison, Wisconsin, US                       | Ph.D                      | 08/2020           | Department of Mathematics                                   |
| University of Wisconsin–Madison, Madison, Wisconsin, US                       | Masters of Arts           | 05/2016           | Department of Mathematics                                   |
| University of Michigan–Ann Arbor, Ann Arbor, Michigan, US                     | Bachelors of Science      | 04/2014           | Department of Mathematics / Department of Political Science |

**Appointments and Positions**

2024 - present Assistant Professor, Dartmouth College, Mathematics, Hanover, New Hampshire, United States

2025 - 2025 Long-Term Visiting Member, Fields Institute, Toronto, Ontario, Canada

2014 - 2020 Teaching Assistant / Research Assistant, University of Wisconsin - Madison, Department of Mathematics, Madison, WI, US

**Products****Products Most Closely Related to the Proposed Project**

1. Madeline Brandt, Juliette Bruce, Melody Chan, Margarida Melo, Gwyneth Moreland, Corey Wolfe. On the top-weight rational cohomology of  $\mathcal{A}_g$ . Geometry & Topology. 2024 March. DOI: 10.48550/ARXIV.2012.02892
2. Bruce J, Erman D. A probabilistic approach to systems of parameters and Noether

normalization. *Algebra & Number Theory*. 2019 December 7; 13(9):2081-2102. Available from: <https://msp.org/ant/2019/13-9/p05.xhtml> DOI: 10.2140/ant.2019.13.2081

3. Juliette Bruce. Asymptotic syzygies for products of projective spaces. *Journal of Algebra*. 2024 July. DOI: 10.1016/j.jalgebra.2024.03.013
4. Juliette Bruce, Lauren Cranton Heller, Mahrud Sayrafi. Characterizing Multigraded Regularity on Products of Projective Spaces. 2021. Available from: <https://arxiv.org/abs/2110.10705> DOI: 10.48550/ARXIV.2110.10705
5. Juliette Bruce, Lauren Cranton Heller, Mahrud Sayrafi. Bounds on Multigraded Regularity. 2022. Available from: <https://arxiv.org/abs/2208.11115> DOI: 10.48550/ARXIV.2208.11115

*Other Significant Products, Whether or Not Related to the Proposed Project*

1. Bruce J, Erman D, Goldstein S, Yang J. Conjectures and Computations about Veronese Syzygies. *Experimental Mathematics*. 2018 June 21; 29(4):398-413. Available from: <https://www.tandfonline.com/doi/full/10.1080/10586458.2018.1474506> DOI: 10.1080/10586458.2018.1474506
2. Juliette Bruce. The quantitative behavior of asymptotic syzygies for Hirzebruch surfaces. *Journal of Commutative Algebra*. 2022 March. DOI: 10.1216/jca.2022.14.19
3. Bruce J, Corey D, Erman D, Goldstein S, Laudone R, Yang J. Syzygies of  $\mathbb{P}^1 \times \mathbb{P}^1$ : Data and conjectures. *Journal of Algebra*. 2022 March; 593:589-621. Available from: <https://linkinghub.elsevier.com/retrieve/pii/S0021869321005160> DOI: 10.1016/j.jalgebra.2021.10.023
4. Ayah Almousa, Juliette Bruce, Michael Loper, Mahrud Sayrafi. The virtual resolutions package for Macaulay2. *Journal of Software for Algebra and Geometry*. 2020 August. DOI: 10.2140/jsag.2020.10.51
5. Eran Assaf, Madeline Brandt, Juliette Bruce, Melody Chan, Raluca Vlad. Tropicalizations of locally symmetric varieties. 2025. Available from: <https://arxiv.org/abs/2505.10504> uri: <https://arxiv.org/abs/2505.10504>

**Certification:**

I certify that the information provided is current, accurate, and complete. This includes but is not limited to information related to domestic and foreign appointments and positions.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Bruce, Juliette in SciENCv on 2025-09-13 12:01:47